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RESEARCH ARTICLE

The Effect of Social Media Load on Academic Performance: A Study on University Students

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What is already known on this topic?

- *Social media overload causes distraction, inability to focus, decreased productivity and has negative effects on academic performance*
- *Information overload has individual effects such as disorganization, loss of time, increased stress and anxiety, difficulty in decision making, decreased productivity, and inability to focus.*
- *Communication overload causes the individual to feel overwhelmed and daily tasks to be interrupted.*
- *Excessive social media load does not have a direct effect on academic performance.*

What this study adds on this topic?

- *Social media overload does not have a direct impact on academic performance.*
- *Communication overload and information overload do not have a direct effect on academic performance.*
- *Social overload negatively affects academic performance.*

Abstract

University students, due to their age and position, use social media extensively in their daily lives. Social media usage has exposed university students to social media overload. In recent years, numerous new studies addressing social media overload and its sub-dimensions have emerged in the international literature. This study investigates the impact of social media overload and its sub-dimensions—communication overload, information overload, and social overload—on academic performance. The study sample consists of students enrolled at Gazi University and Hacı Bayram Veli University. In this context, data were collected from 617 university students through an online survey using a convenience sampling method, employing 7-point Likert-type scales. A quantitative data analysis method was applied for data analysis. Descriptive statistics, demographic distributions, normality analysis, validity and reliability tests, and regression analysis were conducted using the SPSS 21.0 software package. Confirmatory factor analysis was performed using the Lisrel 8.5 software package. The analysis results revealed that communication overload and information overload, as sub-dimensions of social media overload, do not have a direct impact on academic performance. However, social overload was found to negatively affect academic performance.

Keywords: Social media, social media overload, information overload, communication overload, social overload, academic performance.

Introduction

The development of the Internet has radically changed the ways of accessing information and communicating. In this process, platforms called 'social media' have come to the fore. Social media is a new socializing space that allows users to connect, interact, produce, and share content. The fact that the Internet makes it possible to access information quickly, easily, and at low cost has supported the growth of social media and resulted in a communication environment that transcends borders in the digital world.

Social media is an area with users from all age groups. However, it is seen that its use is more widespread, especially among young. University students frequently use digital applications in their daily lives due to their age and location (Shi et al., 2020). University students use social media applications more for reasons such as education, accessing information quickly and easily, exchanging information and ideas, having fun, sharing their feelings and thoughts, establishing social ties, communicating, producing content, making criticism, playing games, reporting location and status, sharing their photos or videos, and creating an agenda. According to a study conducted by the Pew Research Centre (2022), at least 92% of young people are active social media users.

According to some studies in the literature, intensive social media use negatively affects students' academic performance (Boogart, 2006; Cao & Tian, 2020; Kirschner & Karpinski, 2010; Kuss et al., 2013; Owusu-Acheaw & Larson, 2015; Turel & Qahri-Saremi, 2016; Whelan, 2020a). In some other studies, it has been determined that students experience distraction and loss of productivity as multitasking activities such as instant messaging increase during academic tasks (Jacobsen & Forshte, 2011; Levine et al., 2007).

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Although social media undoubtedly provides many advantages to users, it exposes them to excessive information, frequent communication, and too many social demands that exceed their cognitive capacities. This situation causes users to spend more attention, time, and energy and feel overloaded. The overload problem that emerged and developed within the framework of social media applications is a new concept in the literature, which is considered "social media overload" (Cao & Sun, 2018; Shi et al., 2020; Whelan et al., 2020a, b; Yu et al., 2019). Social media overload is experienced when information, communication, and social demands from social media applications exceed a person's ability to cope (Yu et al., 2019). In this context, social media overload is a new problem that has emerged, especially in recent years, due to the widespread use of social media, and has been discussed extensively in the international literature. Social media overload is addressed in three dimensions: communication overload, information overload, and social overload (Cao & Sun, 2018; Shi et al., 2020; Whelan et al., 2020a).

Students exposed to social media overload lose a sense of control over their own time and tasks and experience various psychological and behavioral problems (Yu et al., 2019). In this context, social media overload, which occurs due to the use of social media that penetrates deeply into all areas of life and leads to various negative consequences, is considered an important issue.

A review of the literature indicates that no study has examined social media overload within the national academic literature. Among the sub-dimensions of social media overload, communication overload, and social overload are being investigated for the first time in the national context. In this regard, based on the aforementioned information and existing research in the literature, the objective of this study is to examine the impact of social media overload and its sub-dimensions on academic performance. This study aims to contribute to understanding the effects of social media on students' academic performance and to address the gap in the literature on this subject.

Conceptual Framework

Communication Overload

The concept referred to in English as "communication load" was first introduced Meier (1963) half a century ago. It was Farace et al. (1977) defined the concept of "communication load" as "a situation in which an individual has more information than they can process within a given period, thereby hindering the completion of important tasks." Since the 2000s, due to individuals being exposed to intense and rapid communication demands from information and communication channels, the term "communication overload" has replaced "communication load." Communication overload is defined as the exposure to an excessive number of messages from various communication channels that exceed the recipient's capacity to manage or interpret them (Cho, 2017). This overload represents a cognitive state related to an individual's capacity to process incoming messages (Cho et al., 2011). Communication overload occurs when continuous communication demands cause individuals to feel overwhelmed and disrupt their daily tasks (Karr-Wisniewski & Lu, 2010).

Although communication overload was initially examined in an organizational context (Ballard & Second, 2006), today, it has become a common problem in individual life and social media use. The proliferation of digital technologies such as the Internet, mobile devices, and social media has further deepened the effects of communication overload (Cao & Sun, 2018; Bahri et al., 2019). Especially in recent years, it is considered one of the sub-dimensions of social media overload and has an important place in understanding the difficulties individuals face in the digital world.

Information Overload

Information overload is defined as a situation that exceeds the capacity for information processing (perceiving, using, storing, and transmitting information in a meaningful way) and causes a decrease in decision-making performance due to the limited cognitive capacity of the individual (Eppler & Mengis, 2004; Miller, 1956). Here, "information processing" refers to all processes related to receiving, storing, and transferring information (data), similar to technological data processing (Klausegger et al., 2007). Bawden et al. (1999) define the concept of information overload is defined as a situation in which the level of relevance, quality, benefit, and quantity of the information presented to the individual prevents efficiency in using the information.

Information overload, which is referred to as "Information Overload" in English, is not a new concept but dates back a few centuries ago. The effects of information overload, which require interdisciplinary approaches, have changed in parallel with the changing conditions over time (Batista & Marques, 2018). It has been researched as an important factor in medicine, education, literature, politics, governance, business, news sources, social media use, and online areas (Bawden & Robinson, 2020). While information overload has been addressed as an issue affecting academics and managers in the past, today, it is rapidly increasing its presence in new social areas such as social media applications that have developed as a feature of the information and communication age (Roetzel, 2017).

Bawden and Robinson (2020) state that the factors affecting information overload are too much information, characteristics of information (such as diversity, quality, complexity, and novelty of information), widespread and sent information (push messages/notifications), personal factors, and individual differences (time of day, noise, lack of sleep, competencies or experience).

In Internet-based social spaces, every user can easily create and publish information. It is becoming increasingly difficult to control the increasing flow of information in these areas (Nawaz et al., 2018). In addition to unreliable and unreliable information of uncertain origin in social media spaces, the same information is repeatedly circulated from different news sources (Renjith, 2017). Viral videos on YouTube, WhatsApp messages, e-mails, short messages (SMS), blog posts, countless posts on X, news feeds, and updates on Instagram and Facebook create information overload (Eppler, 2015).

Individual effects of information overload are disorganization, loss of time, increase in stress and anxiety, decrease in self-confidence, difficulties in cognitive capacity, difficulty in decision-making (analysis paralysis), decrease in productivity, focusing problems, difficulty in decision-making and processing, and financial losses (Renjith, 2017). At the organizational level, information overload leads to stress, job dissatisfaction, deterioration of physical health (Edmunds & Morris, 2000), and tension in personal relationships (Kırsh, 2000). Excessive information overload deteriorates the quality of the work produced by employees and prevents employee productivity and effectiveness (Sabeeh & Ismail, 2013).

Social Overload

Another dimension of social media burden, social overload, is based on social support theory (Maier et al., 2012). Social overload is defined as a situation in which an individual perceives that they provide too much social support to other users on social networking sites (Maier et al., 2015a). Social overload causes social media fatigue (Zhang et al., 2016), social media burnout (Maier et al., 2015b), the discontinuous intention of social media users (Cao & Sun, 2018), stress (Fu et al., 2020), and negatively affects academic performance (Yu et al., 2019).

The fact that socialization now takes place in virtual environments rather than physical environments has moved the phenomenon of social overload to online areas. In areas such as Facebook, Instagram, and WhatsApp, and social media users feel responsible for following, liking, commenting, responding to, solving problems, chatting, or entertaining their friends to improve the quality of life of other users they are in contact with. In parallel with social media use, online social support research has increased since 2010. Online friendships in virtual communities have become an important source of social support (Wong & Ma, 2016). Nabi et al. (2013) found that Facebook users felt strong social support in proportion to the number of friends on their lists. However, users feel an excessive social burden as a result of too many social demands from other users that they have difficulty meeting (Maier et al., 2012). With increasing social media use, users may experience social overload when unable to meet the increasing social support demands due to their limited capacity may experience social overload when they are unable to meet (Wang et al., 2020). Staying connected on social media can cause social overload to always exist depending on social interactions (Lin et al., 2020). The interaction on social media, which has become the changing face of socialization, sometimes provides emotional support to individuals but also exposes them to excessive social burdens as the social network expands and interaction increases.

Academic Performance

Performance is French "performance" and its translation into Turkish is "achievement." In the dictionary of the Turkish Language Association, it is defined as "the success shown in putting forward any work, game, work, etc." (Turkish Language Institution, 2021). In addition, while the concept of performance represents an output, it can be associated with many fields such as business, art, sports, and education due to its nature (Çelenlioğlu, 2020, pp. 8–9).

Performance is defined as the endeavor to achieve something with the actions that people exhibit in their lives or working life and all the activities that determine the goals and objectives that develop the responsibilities and visions of each person (Uyargil, 2013, p. 19).

When the literature is examined, it is seen that the concept of academic performance is mostly discussed and defined in the context of academics. In this study, the concept of "academic performance" is evaluated in the context of students. In this context, academic performance can be defined as the qualitative (quality) or quantitative (quantity) results of the actions that students exhibit to fulfill the tasks assigned to them in the education process. Tuckman (1975, p. 34) defines academic performance as "a clear demonstration of a student's understanding, concepts, skills, ideas and knowledge."

For students, academic performance generally refers to the totality of assessment criteria (such as grades and test scores) that can affect student achievement in school, including the student's cognitive skills and attitudes (such as attention, memory, motivation, learning process, and self-confidence), academic behaviors (such as completion of homework/projects and tasks, completion times, and class attendance) and their results (Rasberry et al., 2011).

Students' academic performance is affected by psychological, economic, social, personal, and environmental factors (Singh et al., 2016). Social media use also affects students' academic performance due to its widespread use and dynamic structure (Jacobsen & Forshte, 2011). Students intensively use applications such as Snapchat, Tik Tok, Tik Tok, and Telegram use. As a result of these uses, Facebook, YouTube, WhatsApp, Instagram, and X, for activities such as playing video and online games, watching films, producing and sharing content, having fun, establishing various collaborations, and interacting and communicating affect students' academic and social life (Fu et al., 2021; Junco, 2012).

Methods

This section, which covers the research design and methodology, provides a detailed description of the research model and hypotheses, as well as comprehensive information regarding the research population and sample. Additionally, it includes details on data collection tools, data analysis methods, and validity and reliability analyses. The data for this study were obtained through an online survey conducted between March and April 2021. To conduct this study, approval was obtained from the Ethics Committee for Social and Humanities Research at Kırıkkale University on (Approval no: 2021/01-09, Date: 08.02.2021).

Research Model, Population, and Sample

This study employs a general survey model and is structured as a descriptive design, one of the quantitative research methodologies, in line with its objectives. Descriptive research aims to determine the existing state of a particular subject. Furthermore, the research model is designed to explain the relationships and effects among the identified concepts, as guided by the research objectives and hypotheses.

The key concepts examined in this study include social media overload and its sub-dimensions—communication overload, social overload, and information overload—as well as academic performance. The research model, developed within the framework of the existing literature, is presented in Figure 1.

The hypotheses developed are as follows: the purpose, scope, and research model of the research.

The fact that individuals encounter information, communication, and social demands that exceed their cognitive capacity and that they have difficulty coping with due to social media use creates a perception of overload in individuals. These overloads perceived due to social media use are referred to as "social media overload" (Whelan et al., 2020a). Social media overload is considered in three dimensions. These are social overload, information overload, and communication overload (Yu et al., 2018; Shi et al., 2020). It is observed that social media overload is a new problem that has emerged especially in recent years due to the widespread use of social media and has been discussed extensively in the international literature.

Studies in the literature indicate that social media overload, which arises due to social media usage, affects employees' job performance (Karr-Wisniewski & Lu, 2010) and students' academic performance (Yu et al., 2019; Whelan et al., 2020a). Based on these studies, it is expected that social media overload and its sub-dimensions—communication overload, information overload, and social overload—will impact students' academic performance. Accordingly, the hypotheses and sub-hypotheses to be examined in this study are as follows:

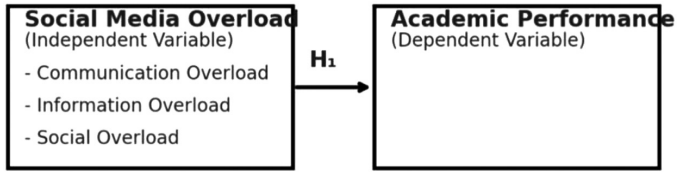


Figure 1.
Research Model.

H_1: Social media overload affects academic performance.

H_{1a}: Communication overload affects academic performance.

H_{1b}: Information overload affects academic performance.

H_{1c}: Social overload affects academic performance.

Population and Sample of the Study

The population of the study consists of university students. Due to time and cost constraints, a non-probability sampling method, specifically the convenience sampling method, was preferred in the research. The study sample was limited to students enrolled at Gazi University and Hacı Bayram Veli University.

During the data collection period, the Spring semester of the 2020–2021 academic year, there were 41,441 registered students at Gazi University and 27,369 registered students at Hacı Bayram Veli University. Between March and April 2021, a survey link was sent to the students in the sample group via email. A total of 630 responses were received from the participants, of which 13 were deemed invalid. Consequently, the analyses were conducted using 617 valid responses.

In this context, Hair et al. (2010) criteria were used to determine the required sample size for the population. According to these criteria, the minimum required sample size should be at least five times the total number of items in the scales used in the research, while the ideal sample size should be ten times the number of items. Since the measurement instruments used in the study contain a total of 19 items, Hair et al. (2010) suggest that a sample size between 95 (19 × 5) and 190 (19 × 10) is sufficient to represent the population. Demographic information about the sample is presented in Table 1.

Data Collection Tools and Analysis Method

The structure of the questionnaire used in the research consists of five questions, including demographic and descriptive information, a social media overload scale, and an academic performance scale. The research scales are valid and reliable scales used in previous studies.

Social media overload consists of three dimensions: communication overload, social overload, and information overload. In the study, the information overload scale was originally developed by Karr-Wisniewski and Lu (2010) and adapted from Yu et al. (2019). This scale consists of three items. The communication overload scale, originally developed by Karr-Wisniewski and Lu (2010), was adapted from the study by Whelan et al. (2020a) and consists of four items. The social overload scale, originally developed by Maier et al. (2015a) and consisting of five items, was obtained from the study by Yu et al. (2019). The academic performance scale, originally developed by Welbourne et al. (1998) and used in the study by Whelan et al. (2020a), consists of seven items.

In this context, to ensure content validity, the initial translation of all scale items from the original language into Turkish was conducted by two English instructors. The back-translation was then performed by a faculty member specializing in the field of business studies. Subsequently, to assess the comprehensibility of the scale in terms of language, it was administered to a group of 12 students, and a few modifications were made. Through these procedures, potential semantic losses during the translation process were examined, and the final version of the scale was determined. All items in the scale were structured using a seven-point Likert scale.

A quantitative data analysis method was employed in the analysis of the data. Within the scope of the analysis, demographic distributions, reliability analysis, correlation analysis, and regression analysis were conducted using the SPSS 21.0 software package. Additionally, confirmatory factor analysis was performed using the Lisrel 8.5 software package.

Table 1.
Demographic Information of the Sample

		Number	Rate (%)
Gender	Male	335	54.4
	Woman	281	45.6
Your class	Preparation	30	4.9
	First class	150	24.3
	Second class	218	35.3
	Third class	115	18.6
	Grade 4	87	14.1
	Other	17	2.8
How many hours a day do you use the Internet on average?	Less than 1 hour	8	1.3
	1–2 hours	14	2.3
	3–4 hours	151	24.5
	5–6 hours	205	33.2
	7 hours and over	239	38.7

Validity and Reliability of Data Collection Tools

Based on the data obtained from the study, the normality distribution of the data was first tested. Subsequently, comparisons were made between the variables. Due to the sample size of the study exceeding 30 ($n = 617$), the Kolmogorov-Smirnov test was employed. The results are presented in Table 2.

When Table 2 is examined, it is determined that the Skewness-Kurtosis values of the dimensions fall within the range of −2 to +2. The fact that these values remain within the specified limits indicates that the data follow a normal distribution (Eraslan et al., 2017; Salam, 2017). Accordingly, the data were assumed to be normally distributed, and it was deemed appropriate to conduct parametric analyses in the study. To determine the reliability level of the scales, the Cronbach's Alpha coefficient was utilized. As a result of the factor analyses, items with a factor loading below 0.30 and items loading onto multiple factors were excluded from the analysis.

The reliability analysis results of the scales in the study are in Table 3:

When Table 3 is examined, it is observed that the reliability levels of the scales used in the study are at an acceptable or good level in terms of reliability tests (Cronbach, 1951). According to Table 3, an examination of the sub-dimensions, the fact that these values are greater than 0.5 indicates that the sub-factors of the scale are also sufficiently reliable (Saruhan & Özdemirci, 2016; Taber, 2018).

As a result of the item-total correlation analysis, it was observed that the item-total correlations of the items forming the sub-dimensions of the social media overload variable—namely, social overload, information overload, and communication overload—ranged between 0.390 and 0.767. In terms of the academic performance variable, the item-total correlations were found to range between 0.605 and 0.730. It was decided that no items would be removed from the scales.

In the next stage, Confirmatory Factor Analysis (CFA) was conducted using the Lisrel 8.5 program to determine the construct validity of each scale.

According to the CFA results for the social media overload scale presented in Table 4, the goodness-of-fit values were as follows: RMSEA (0.073); χ^2/SD (4.306); SRMR (0.058); GFI (0.94), indicating

Table 2.
Descriptive and Normality Distribution Statistics of Scale Dimensions

	Minimum	Maximum	Centre	SS	Skewness	Kurtosis
Social media overload	1.00	7.00	3.8683	1.04682	.014	−.095
Information overload	1.00	7.00	4.4419	1.37421	−.220	−.569
Communication overload	1.00	7.00	4.1933	1.32273	−.154	−.417
Social overload	1.00	7.00	3.2642	1.42356	.381	−.544
Academic performance	1.00	7.00	5.3066	1.09620	−.973	1.540
N			617			

Note: SS = standard deviation; KS = Kolmogorov–Simirnov, *$p < .05$.

an acceptable level of fit; and NFI (0.95), AGFI (0.91), and CFI (0.96), indicating a good level of fit. Accordingly, it can be stated that the construct validity of the social media overload scale is at an acceptable level (Table 5).

In the examination of standardized factor loadings, it was accepted that the loading values should be at least 0.30 or higher (Büyüköztürk, 2013).

According to the measurement model presented in Figure 2, the standardized regression coefficients, or factor loadings, for the paths on the unidirectional arrows, as well as the items of the measurement model confirmed with three factors and consisting of 12 items, are displayed. An examination of the goodness-of-fit values and factor loadings in the figure indicates that the goodness-of-fit values are at a good fit level and that the factor loadings exhibit complete alignment.

According to the CFA results for the academic performance scale, the goodness-of-fit values were as follows: RMSEA (0.030), χ^2/s.d. (1.55), SRMR (0.006), AGFI (0.98), GFI (0.99), CFI (0.99), and NFI (0.99), all of which indicate a good level of fit.

Among the tested models, items 2 and 3 of the academic performance scale were determined to disrupt the model's goodness-of-fit values, and thus, it was decided to remove these items from the model. The resulting PATH diagram is presented below.

According to the measurement model in Figure 3, the items included in the measurement model, which was validated with a single factor consisting of five items, as well as the factor-confirmed model and the factor loadings of these items, are displayed. When examining the standardized factor loadings, it was considered that the values should be at least 0.30 or higher. An assessment of the goodness-of-fit values and factor loadings presented in Figure 3 confirms that the goodness-of-fit values indicate a good level of fit and that the factor loadings exhibit complete alignment.

Accordingly, the results of the Cronbach's Alpha analysis, conducted to test the reliability of the newly obtained factor structures, are presented below. An examination Table 6 reveals that the reliability levels of the dimensions are at an acceptable to high level. Additionally, the item-total correlation analysis results indicate that the item-total correlations for each dimension range between 0.29 and 0.763.

Table 3.
Reliability Analysis Results

Dimensions	Article Number	Cronbach's Alpha
Sub-dimensions		
Information overload	3	.761
Communication overload	4	.719
Social overload	5	.870
Upper dimensions		
Academic performance	7	.880
Social media overload	12	.838

Findings

In this section, the variables' correlation analysis was applied. To determine the relationship between Afterwards the hypotheses, the findings are presented. Regression applied to analysis test

Relationship Between (Correlation) Independent and Dependent Variables

The results of the correlation analysis applied to test the relationship between academic performance and social media overload and its sub-dimensions, information overload, social overload, and communication overload, are in Table 7.

When Table 7 is examined, no statistically significant relationship was found between academic performance and social media overload variables according to the Pearson correlation analysis method ($p > .05$). However, the correlation coefficient between academic performance and the social overload dimension was found to be significant ($p < .05$). When the correlation coefficient between academic performance and social overload is analyzed, it can be stated that there is a negative but weak relationship between academic performance and social overload ($r = -.112$). This finding indicates that as social overload increases, academic performance decreases. In other words, as social overload decreases, academic performance increases. No statistically significant relationship was found between academic performance and the other sub-dimensions of social media overload.

In line with the research model and hypotheses, the effect of social media load on academic performance is presented in Table 8

As can be seen in Table 8, when the regression model created to investigate the effect of social media overload on academic performance is examined, the effect of participants' perceptions of social media overload on academic performance is not statistically significant

Table 4.
Social Media Overload Scale CFA Goodness of Fit Values

Compliance parameter	Fit parameter statistic	Acceptable fit values	Good fit values
RMSEA	.073	$0.05 \leq RMSEA \leq 0.08$	$0 \leq RMSEA \leq 0.05$
χ^2/SD	4.306	$2 \leq \chi^2/SD \leq 5$	$0 \leq \chi^2/SD \leq 2$
SRMR	0.058	$0.05 \leq SRMR \leq 0.10$	$0 \leq SRMR \leq 0.05$
NFI	0.95	$0.90 \leq GFI \leq 0.95$	$0.95 \leq GFI \leq 1.00$
GFI	0.94	$0.85 \leq GFI \leq 0.95$	$0.95 \leq GFI \leq 1.00$
AGFI	0.91	$0.80 \leq AGFI \leq 0.90$	$0.90 \leq AGFI \leq 1.00$
CFI	0.96	$0.90 \leq CFI \leq 0.95$	$0.95 \leq CFI \leq 1.00$

AGFI, adjusted goodness of fit index; CFI, comparative fit index; GFI, goodness of fit index; NFI, normed fit index; RMSEA, root mean square error of approximation.

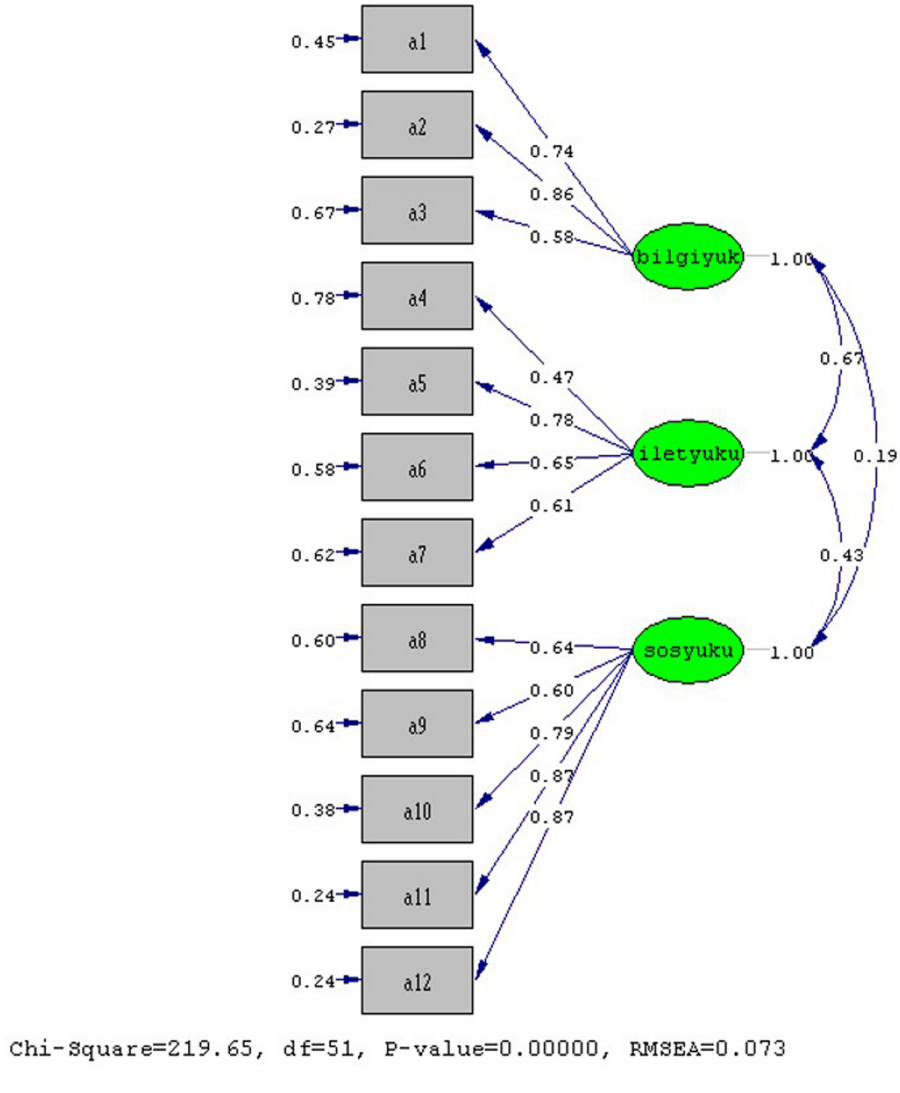


Figure 2.
Confirmatory Factor Analysis PATH Graph for Social Media Overload Scale.

(it can be said that > 0.05). Accordingly, it is observed that the hypothesis "H_1*: Social media overload affects academic performance"* is not statistically supported.

In line with the research model and hypotheses, the effects of social media overload sub-dimensions: communication overload, information overload, and social overload on academic performance are given in Table 9.

Table 5.
Academic Performance Scale CFA Goodness of Fit Values.

Compliance Parameter	Fit Parameter Statistic	Acceptable Fit Values	Good Fit Values
RMSEA	0,030	0.05 ≤ RMSEA ≤ 0.08	0 ≤ RMSEA ≤ 0.05
χ^2/SD	1,55	2≤ χ^2 ≤ 5	0≤ χ^2 ≤ 2
SRMR	0,006	0.05 ≤ SRMR ≤ 0.10	0 ≤ SRMR ≤ 0.05
NFI	0,99	0.90 ≤ GFI ≤ 0.95	0.95 ≤ GFI ≤ 1.00
GFI	0,99	0.85 ≤ GFI ≤ 0.95	0.95 ≤ GFI ≤ 1.00
AGFI	0,98	0.80 ≤ AGFI ≤ 0.90	0.90 ≤ AGFI ≤ 1.00
CFI	0,99	0.90 ≤ CFI ≤ 0.95	0.95 ≤ CFI ≤ 1.00

AGFI, adjusted goodness of fit index; CFI, comparative fit index; GFI, goodness of fit index; NFI, normed fit index; RMSEA, root mean square error of approximation.

According to Table 9, the regression analysis examining the effect of social media overload sub-dimensions on the academic performance variable, the established model is significant ($p < .05$). However, when examining the coefficient of determination, it is observed that the explained variance ratio is low ($R^2 = .014$; proportionally: 1.4%). Based on the coefficients, only the social overload dimension is found to have a statistically significant effect on academic performance ($p < 0.05$). The negative slope of the coefficient (β=−0.132) suggests that social overload negatively affects academic performance. Accordingly, the "H_{1c}*: Social overload affects academic performance."* hypothesis is statistically supported, while the "H_{1a}*: Communication overload affects academic performance."* and "H_{1b}*: Information overload affects academic performance."* hypotheses are not statistically supported.

When the model goodness-of-fit values are analyzed, it is seen that VIF, Tolerance, and Durbin Watson statistics indicate that there is no multicollinearity problem in the model.

Discussion, Conclusion, and Recommendations

This study aims to examine the impact of social media overload and its sub-dimensions—information overload, social overload, and communication overload—on the academic performance of university

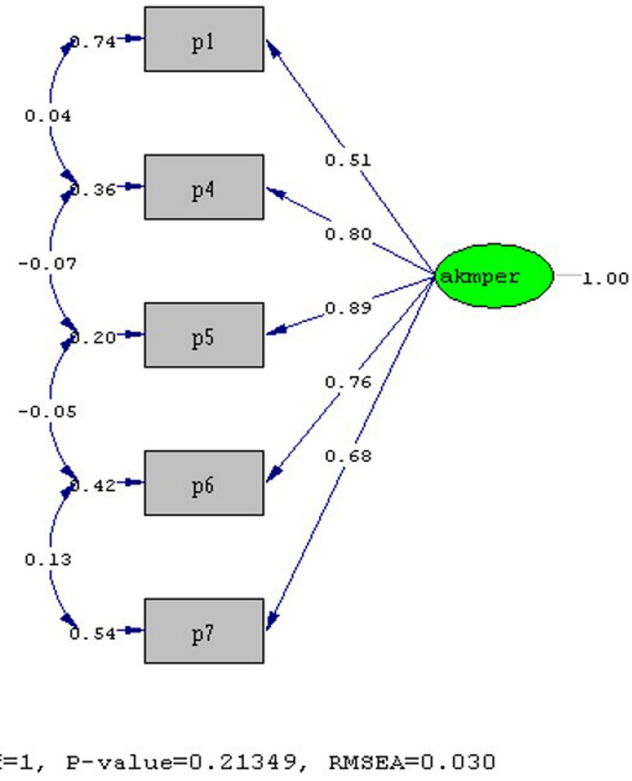


Figure 3.
Confirmatory Factor Analysi PATH Graph for Academic Performance Scale.

students. The study includes 617 students enrolled in two public universities in Ankara. The first section of the study provides information on the concepts of social media overload and its sub-dimensions, namely information overload, social overload, communication overload, and academic performance. The second section focuses on the analysis of the data obtained within the scope of the study.

Table 6.
Reliability Analysis Results According to the New Factor Structure

Dimensions	Article Number	Cronbach's Alpha
Sub-dimensions		
Information overload	3	.761
Communication overload	4	.719
Social overload	5	.870
Upper dimensions		
Academic performance	5	.842

In the first section of the study, the suitability of the scales used to the structure defined in the literature was tested. To assess the construct validity of the scales, confirmatory factor analysis was conducted. The results of this analysis indicated that the goodness-of-fit values of the social media overload and its sub-dimensions—information overload, social overload, communication overload—as well as the academic performance scale, were at a good fit level, and the factor loadings were observed to be in full compliance.

In order to test the research model and hypotheses developed within the framework of the literature, a regression analysis was conducted. The results of the analysis revealed that social media overload and its sub-dimensions, information overload, and communication overload, do not have a direct impact on academic performance. This may be attributed to the possibility that students engage in social media activities during their free or leisure time. Another possible reason is that the effects may vary depending on the students' purposes

Table 7.
The Relationship Between Social Media Overload and Its Sub-dimensions and Academic Performance

		Information Overload	Communication Overload	Social Overload	Social Media Overload	Academic Performance
Information overload	*r*	1				
	p	-				
Communication overload	*r*	.535**	1			
	p	.000				
Social overload	*r*	.188**	.368**	1		
	p	.000	.000			
	p	.135	.042	.000		
Social media overload	*r*	.660**	.806**	.784**	1	
	p	.000	.000	.000		
Academic performance	*r*	.052	.006	−.112**	−.044	1
	p	.200	.885	.005	.272	-

H_0: $\Upsilon_k = 0$ (The correlation coefficient is not significant.)

Table 8.
Results of Regression Analysis Between Social Media Overload and Academic Performance

		Model Significance				Coefficient	
Independent Variable	**Dependent Variable**	**F**	**P**	**R**	**R²**	β	**p**
Social Media Overload	Academic Performance	1,208	0,272	0,044	0,002	–0,044	0,272

Table 9.
Regression Analysis Results Between the Sub-dimensions of Social Media Overload and Academic Performance

		Model Significance			Coefficient				Durbin
Independent Variable	**Dependent Variable**	***F***	***P***	***R²***	*B*	***P***	**VIF**	**Tolerance**	**Watson**
Information Overload	Academic Performance	3.826	.010	0.014	.066	.161	1.402	0.713	1.766
Communication Overload					.019	.705	1.564	0.639	
Social Overload					−.132	.002*	1.157	0.864	

*$p < .05$; R^{2} adjusted coefficient of determination.

for using social media. Hameed et al. (2022) found that the effect of social media use varies depending on the purpose for which students engage with social media. Specifically, while using social media for academic purposes has a positive relationship with academic performance, using social media for non-academic purposes is negatively associated with students' academic performance. When examining previous studies, some research findings align with the results of this study, indicating no significant relationship between social media use and students' academic performance (Alwagait et al., 2015; Rashid & Asghar, 2016). However, in contrast to these findings, other studies have identified the negative effects of social media overload, information overload, and communication overload on academic performance through mechanisms such as inadequate self-regulation (Whelan et al., 2020a), technostress (Yu et al., 2019), or emotional exhaustion (Shi et al., 2020).

According to another finding of the study, social overload negatively affects academic performance. The result obtained regarding social overload aligns with the existing literature (Shi et al., 2020; Whelan et al., 2020a; Yu et al., 2019). Several studies have observed the behavioral and psychological negative consequences of social overload on social media users (Cao & Sun, 2018; Maier et al., 2012, 2015a, b). The reason why exposure to social overload negatively impacts the academic performance of university students is thought to be the overwhelming number of social demands they receive from their connections on social media. This situation leads social media users to lose their sense of control over their time and responsibilities, causing them to feel overloaded. Students' engagement with these social demands may result in dedicating less time to academic work, decreased focus, or reduced productivity. Feng et al. (2019) found that students who frequently use Facebook tend to be more distracted from academic tasks and have lower grade point averages. Additionally, Busalim et al. (2019) observed that the frequency of Facebook use has a significant negative impact on students' academic performance and that Facebook addiction is a positive predictor of this effect (Cao & Tian, 2020).

Social media users are social individuals who need social interaction and frequently seek it, engaging with their communities in this manner (Junglas et al., 2013). Platforms characterized by intense social interactions, such as Facebook, Instagram, and WhatsApp, compel their users to make continuous efforts to respond to their friends. In the current state of communication, the prevailing notion is that individuals must always be available for interaction. As a result, people can request social assistance from their connected friends via social media without any time or space constraints. University students also struggle to refuse such requests, spending a significant portion of their time responding to incoming demands. Additionally, social requests received through social media can distract university students, make it difficult for them to concentrate, or cause them to feel overwhelmed. For these reasons, it appears that when university students perceive social overload through social media, they may experience negative effects on their academic performance. University education is a period during which students not only fulfill various academic responsibilities but also prepare for their professional careers. This situation is also expected to negatively impact career development performance, which is a crucial factor in students' transition to the workforce.

Today, social media has become an integral part of life and an inevitable tool for university students, who constitute a significant portion of internet users referred to as "digital natives" (Gümüş, 2021, p. 182). However, evidence in the literature indicates that research on this subject remains insufficient, and the findings from previous studies are notably inconsistent and inconclusive.

In conclusion, our study provides significant insights into the impact mechanism of social media-induced overloads on university students' academic performance, contributing to a better understanding of the negative aspects of social media use. Previous research has demonstrated that students' work experience and age play a crucial role in their use of social networks (Benson & Filippaios, 2015). Therefore, it is recommended that future studies examine this research topic across students at different levels of education. This study is significant in terms of its contribution to the literature. Moreover, the findings are considered important for defining the boundaries and conditions of the relevant theory. It is suggested that future research should explore the underlying reasons for the results presented in this study.

Moreover, digital technologies such as artificial intelligence, cloud computing, big data, and smartphones—emerging with the advancement of the internet—have influenced various fields, including communication, education, marketing, and finance, bringing about fundamental changes and transformations in both personal and professional life (Özdemir & Çetiner, 2021; Yüzbaşıoğlu, 2023). These digital technologies, which have become a necessity in modern life, are expected to increase overload. In particular, it is recommended that the phenomena of information overload and communication overload, as well as strategies for coping with them, be examined across all digitalized domains.

Data Availability Statement: The data that support the findings of this study are available on request from the corresponding author.

Ethics Committee Approval: Ethics committee approval was received for this study from the Ethics Committee of Kırıkkale University (Approval no: 2021/01-09, Date, 08.02.2021).

Informed Consent: Written informed consent was obtained from participants who participated in this study.

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